Lesson 8.4 Guided Notes

Session 1 · Writing Inequalities

I can write an inequality that describes a constraint in a real situation. · 6.EE.B.8

Word / Spanish	Meaning — complete the blank	Example
inequality / desigualdad	A math sentence that _____ using <, >, ≤ or ≥.	x ≥ 18
at least / al menos	That number _____ (≥).	at least 5: x ≥ 5
at most / a lo sumo	That number or _____ (≤).	at most 8: x ≤ 8
more than / más que	_____ than the number, not equal to it (>).	more than 5: x > 5

Key idea

First decide whether the boundary number is _____.

Allowed: use ≤ or _____. Not allowed: use < or _____.

Worked example

To make the team, a swimmer must swim more than 20 laps. Write an inequality for L, the laps.

Key words: "_____."

Is 20 allowed? _____, so use >.

Inequality: L _____ 20.

Answer: L > 20, so 20 laps is _____ a solution.

You try

1. Write an inequality for each phrase. Use x.
a. at least 13 _____
b. fewer than 9 _____

_____

2. A coupon works on orders of $50 or less. Write an inequality for d, the order amount. Does a $50 order work?

_____

Sum it up: To choose an inequality symbol, I ___ because ___.

_____

Session 2 · Representing Inequalities

I can represent an inequality with a set of solutions and describe them. · 6.EE.B.8

Word / Spanish	Meaning — complete the blank	Example
open circle / círculo abierto	The boundary is _____ a solution (< or >).	x > 3: open circle at 3
closed circle / círculo cerrado	The boundary _____ a solution (≤ or ≥).	x ≤ 3: closed circle at 3
solutions of an inequality / soluciones de una desigualdad	_____ the numbers that make it true.	x > 3: 3.5, 4, 10, …

Key idea

Draw a circle at the _____ number.

For < or ≤, the arrow points _____. For > or ≥, it points _____.

Worked example

Graph x < 3.

The boundary is _____.

< does not include 3, so draw an _____ circle.

Draw the arrow to the _____.

Answer: Sample solutions: 2, 0, −1.5

You try

1. Write the inequality shown. Use x.



_____

2. Is each number a solution of n > 7? Write yes or no.
a. 7 _____
b. 7.5 _____

_____

Sum it up: To graph an inequality, I ___ because ___.

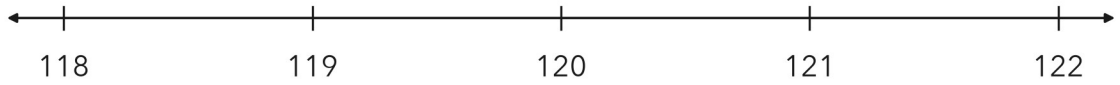
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Session 1 · Model, reason, and explain

Writing Inequalities · 6.EE.B.8

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A ride requires height at least 120 cm. Write a constraint and test 119,120,121. Explain whether the boundary is included.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: confuses strict and inclusive boundaries.. Explain what changes and what stays true.

_____

_____

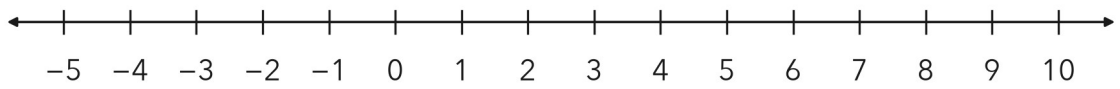
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Representing Inequalities · 6.EE.B.8

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A club needs more than 5 members to run an event. Write an inequality and a number-line representation. Are 5.5 members feasible even though 5.5>5?



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: treats every real-number solution as feasible for a count context.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?